SOLUTION ARCHITECTURE DOCUMENT

Secure Multi-Agent Support System

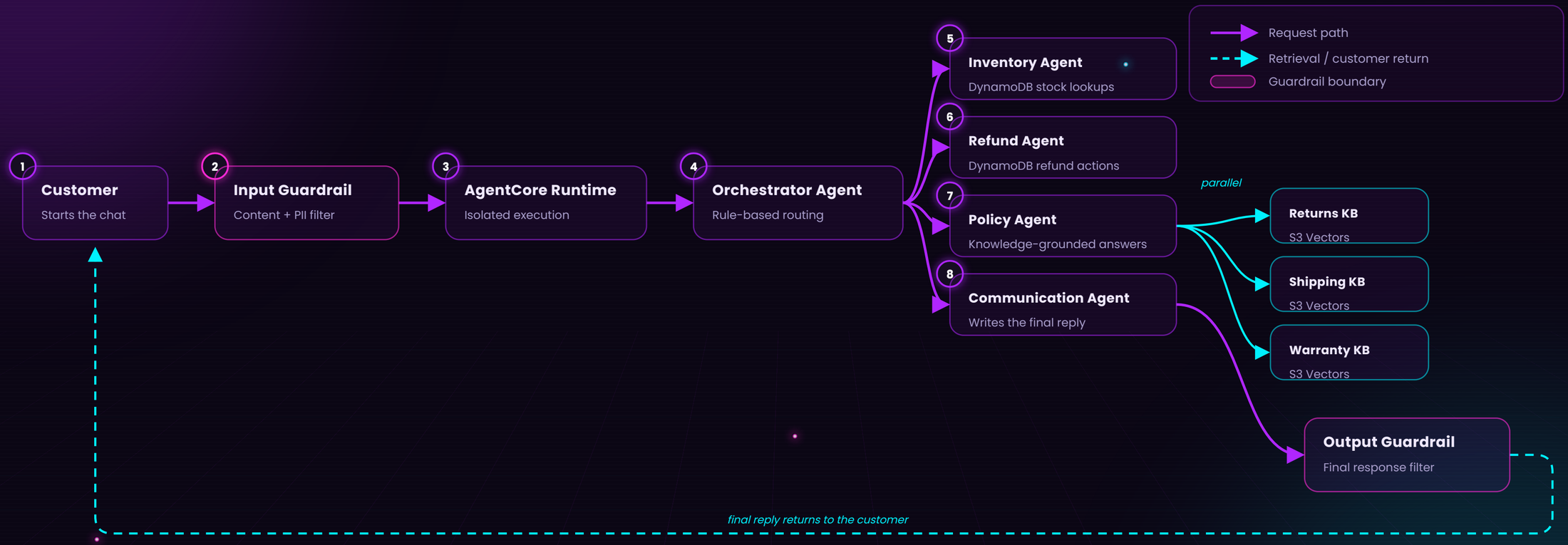
Amazon Bedrock AgentCore | Strands Agents | Knowledge Bases

SPECIALIST AGENTS. **ONE SAFETY LAYER.** FULL VISIBILITY.

Designed by **Dr. Mfon Nta** — Cloud Security Engineer · AI Agents

Lovable

End-to-End Architecture



Workflow State

DynamoDB · optimistic locking · 24h TTL

STATE

AgentCore Memory

Session summaries · 7 days

MEMORY

Observability

CloudWatch Logs + X-Ray traces

TRACING

Lovable

Three Requests, Three Paths

LANE 01

Return request

Tools used: Inventory → Refund

Orchestrator → Inventory Agent → Refund Agent → Communication

LANE 02

Policy question

Knowledge retrieval

Orchestrator → Policy Agent → (3 parallel retrievers: Returns KB, Shipping KB, Warranty KB) → Communication

LANE 03

Math question

No tools needed

Orchestrator → Communication — *answers directly, zero tool calls*

Routing is decided by **explicit rules**, not left to chance.

Lovable

Trust is designed in, not bolted on.

01

Content filters

Both guardrails screen every message in — and every reply out.

02

Personal data

Card and ID numbers blocked; emails and phones masked.

03

Off-limits topics

Competitors, price haggling and legal threats are refused.

04

Least privilege

One execution role, scoped to only what each agent needs.

05

Session isolation

Every conversation is sandboxed and expires automatically.

06

Deterministic routing

Explicit rules choose the agent — never model guesswork.

07

Retries & fallbacks

Graceful fallbacks keep the session alive when a tool fails.

08

Audit trail

Every decision and tool call is logged, end to end.

Lovable